

AN14051

An Approach to Enable a Glitch Filter on FlexIO

Rev. 1 — 13 October 2023

Application note

Document Information

Information	Content
Keywords	FlexIO, Glitch, Filter, i.MX RT
Abstract	FlexIO modules are widely used in the i.MX RT series MCUs, which provide a highly flexible configuration method that can simulate many different communication protocols, such as UART, I ² C, SPI, and I ² S.



1 Introduction

FlexIO modules are widely used in the i.MX RT series MCUs, which provide a highly flexible configuration method that can simulate many different communication protocols such as UART, I²C, SPI, and I²S.

The hardware is based on RT1010-EVK, and the software is based on SDK2.13.0.

2 Reason why FlexIO needs glitch filtering

For example, using FlexIO to simulate a UART can theoretically support a high baud rate, but only sample once per bit. A single sample causes erroneous data to be received due to the presence of glitches. In this case, a glitch filter can be designed with an additional timer to avoid errors.

3 Basic principle of glitch filtering using a timer

Supposing that there is a square wave signal with a frequency of 2 Hz and a timer clock frequency of 1000 Hz, its waveform is as shown in [Figure 1](#).

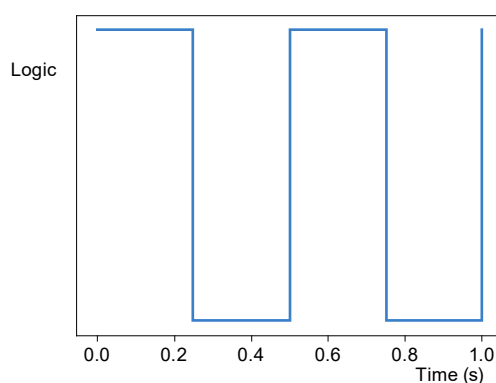


Figure 1. Square wave waveform

In signal transmission scenarios, due to interference, a series of glitches is generated, which can cause incorrect transmission values during sampling only once per bit. The square wave signal waveform with random glitches is shown in [Figure 2](#), where some glitches are less than 5 clock cycles in length and some glitches are longer than 15 clock cycles.

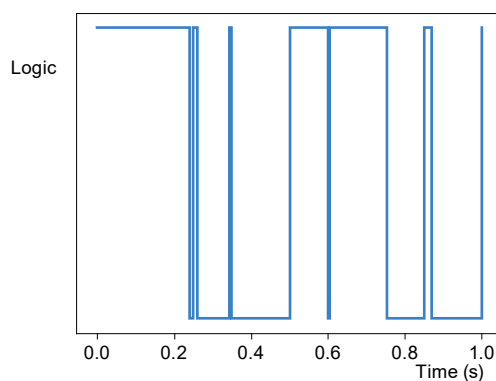


Figure 2. Square wave waveform with random glitches

To eliminate these glitches, enable a timer in FlexIO and set the control register `TIMCFG` of the timer as follows:

- The timer operates in 16-bit mode.
- The trigger source selection for the timer comes from the input pin.
- The pin of the timer is configured as output.
- The timer is always enabled and never turned off (`TIMENA=0`, `TIMDIS=0`).
- When the input and output of the timer are equal, the timer is in the reset state (`TIMRST = 0x3`).
- The output of the timer is not affected by the reset pin, and the initial state can be 0 or 1 (`TIMOUT = 0` or `1`).
- The initial count value of the timer is equal to `filter_width` minus 1 (`TIMCMP = filter_width - 1`).
- The `filter_width` is the number of cycles in which the input signal is continuously stable to update the output signal, and its value must be larger than 1.

By setting different initial count values, the timer can be used to filter high-frequency glitches in the signal, while also introducing a time delay of the filtered signal with the length of `filter_width * clock cycles`. When the initial count equals to 10, the initial signal and the filtered signal are shown in [Figure 3](#).

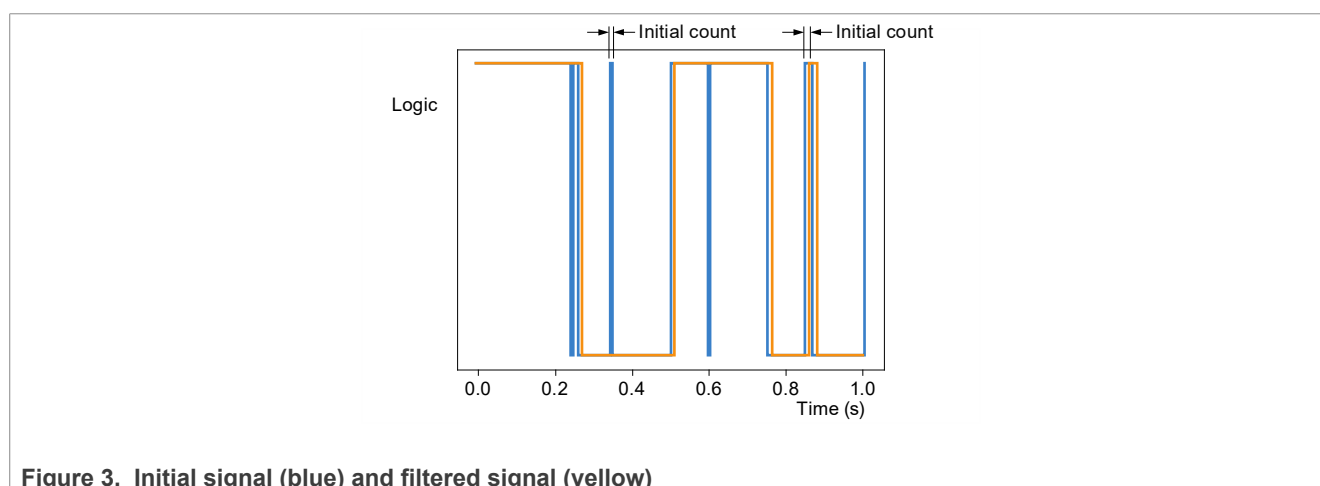


Figure 3. Initial signal (blue) and filtered signal (yellow)

As shown in [Figure 3](#), when the glitch width is greater than `filter_width * clock cycles`, it is not filtered. The glitch is filtered only when the glitch width is less than `filter_width * clock cycles`.

The timer can complete the filtering because when the signal toggles, the counting value decreases from the initial value. It means that the input state of the timer is not the same as the output state, and the timer exits the reset state to start counting. When the 16-bit counter equals 0 and decrements, the timer output switches and the counter is reloaded from the comparison register. Therefore, if the toggled signal is held for less than `filter_width * clock cycles`, the count value is reset before it drops to 0, the output signal remains at the original level, and the short-term signal toggle is filtered.

FlexIO pins are internally connected within the same instance, the output pin of the timer used as a glitch filter is internally connected to other shifters and timers even if that pin(`IOMUXC`) is not enabled at the device level.

4 Filter the square wave signal output from GPIO on RT1010

On the i.MX RT1010 chip, a square wave signal can be output using a GPIO and input to the FlexIO timer.

- The `GPIO_08` is configured as a square wave generator.
- The `GPIO_11` is configured as an input pin of the FlexIO timer.
- The `GPIO_12` is configured as an output pin of the FlexIO timer.

The timer of the FlexIO can be configured to the `TIMCFG` register to change the response of the timer to the input signal.

Table 1. TIMCFG-TIMOUT

25-24 TIMOUT	Timer Output Configures the initial state of the Timer Output and whether it is affected by the Timer reset. • 00b - Timer output is logic one when enabled and is not affected by the Timer reset. • 01b - Timer output is logic zero when enabled and is not affected by the Timer reset. • 10b - Timer output is logic one when enabled and on the Timer reset. • 11b - Timer output is logic zero when enabled and on the Timer reset.
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By configuring the TIMOUT segment to 00b, the initial output value of the timer can be set to 1 and the output value remains unchanged when reset.

Table 2. TIMCFG-TIMDEC

21-20 TIMDEC	Timer Decrement Configures the source of the Timer decrement and the source of the Shift clock. • 00b - Decrement counter on FlexIO clock. Shift clock equals Timer output. • 01b - Decrement counter on Trigger input (both edges). Shift clock equals Timer output. • 10b - Decrement counter on Pin input (both edges). Shift clock equals Pi input. • 11b - Decrement counter on Trigger input (both edges). Shift clock equals Timer output.
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Configure the TIMDEC segment to 00b, which can decrement the count value every clock cycle.

Table 3. TIMCFG-TIMRST

18-16 TIMRST	Timer Reset Configure the condition that causes the tier counter (and optionally the tier output) to be reset. In 8-bit counter mode, the timer reset only resets the lower 8-bit that configure the baud rate. In all other modes, the timer reset resets the full 16 bits of the counter. • 000b - Timer never resets. • 001b - Reserved. • 010b - Timer reset on Timer Pin equal yo Timer Output. • 011b - Timer output on Timer Trigger equal to Timer Output. • 100b - Timer reset on Timer Pin rising edge. • 101b - Reserved. • 100b - Timer reset on Trigger rising edge. • 111b - Timer reset on Trigger rising or falling edge.
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Configuring the TIMRST to 11b allows the count value to be reset when the input value equals the output value.

Configure the FlexIO timer as follows in C code:

```

timerConfig.triggerSelect = FLEXIO_TIMER_TRIGGER_SEL_PININPUT(3);
timerConfig.triggerPolarity = kFLEXIO_TimerTriggerPolarityActiveLow;
timerConfig.triggerSource = kFLEXIO_TimerTriggerSourceInternal;
timerConfig.pinConfig = kFLEXIO_PinConfigOutput;
timerConfig.pinSelect = 4;
timerConfig.pinPolarity = kFLEXIO_PinActiveLow;
timerConfig.timerMode = kFLEXIO_TimerModeSingle16Bit;
timerConfig.timerOutput = kFLEXIO_TimerOutputOneNotAffectedByReset;
timerConfig.timerDecrement = kFLEXIO_TimerDecSrcOnFlexIOClockShiftTimerOutput;
timerConfig.timerReset = kFLEXIO_TimerResetOnTimerTriggerEqualToTimerOutput;
timerConfig.timerDisable = kFLEXIO_TimerDisableNever;
timerConfig.timerEnable = kFLEXIO_TimerEnabledAlways;
timerConfig.timerStop = kFLEXIO_TimerStopBitEnableOnTimerDisable;
timerConfig.timerStart = kFLEXIO_TimerStartBitEnabled;

timerConfig.timerCompare = 0xFFFF;

```

Figure 4. FlexIO timer configuration

As described above, the timer configuration conditions mentioned in [Section 3](#) can be implemented so that the timer operates as a glitch filter.

5 Revision history

[Table 4](#) summarizes the revisions to this document.

Table 4. Revision history

Revision number	Release date	Description
1	13 October 2023	Initial public release

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